Team Name: _____ Team Number: _____

Part 1: Crosses, Pedigrees, & Karyotypes

General Knowledge

1. ___100%___
2. ___1/64___
3. ___c___
4. ___b___
5. ___a, c___ (0.2 pts per option correct)
6. ___a___
7. ___d___
8. ___X^AX^a___ (capitalization matters)
9. ___a___
10. ___b___

Investigation: Pass the Peas

11. ___Yy___ (capitalization matters)
12. ___96___
13. ___192___
14. __Green & blue__ (½ pt for g, ½ for b)
15. ___c___
16. _enzymes D & E_ (½ pt for g, ½ for b)
17. ___pp___ (capitalization matters)
18. ___PP___ (capitalization matters)
19. ___3/16___
20. ___72____

Challenge Problems

21. ___0.72 or 72%___
22. ___7___
23. ___2:1___ (ratio must be in this order)
24. ___1/2 or 0.5___
25. ___c___

Part 2: DNA Structure & Replication

General Knowledge

1. ___deoxyribonucleic acid___
2. ___b___
3. ___a___
4. ___uracil___
5. ___b___
6. ___5' end___
7. ___guanine___
8. ___purine___
9. ___d___
10. ___b___

Investigation: *C. urchinis*

11. ___18%___
12. ___36%___
13. ___Chargaff___
14. ___14%___
15. __36%___
16. ___16 million___
17. ___2.78 ± 0.02____ hours
18. ___11.11 ± 0.02____ hours
19. ___40,000___
20. ___3.22 ± 0.05____ hours

Challenge Problems

21. _a_ , _b_ , _d_ , _c_ (1/3 pt per blank)
22. I. ___c___ II. ___d___
23. ___conservative___ model
24. Test Tube __C__
25. ___DNA ligase___

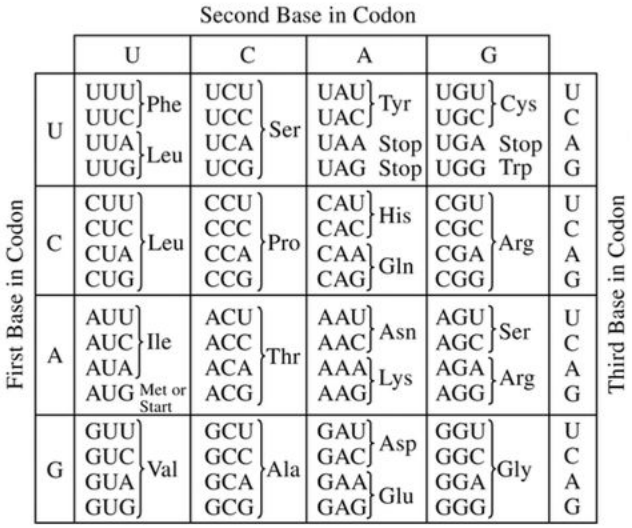
Part 3: Mitosis & Meiosis

General Knowledge

1. ___a___
2. ___c___
3. ___c___
4. ___a___
5. ___d___
6. ___c___
7. ___a___
8. ___c___
9. ___b___
10. ___a___

Challenge Problems

11. __900__ polar bodies
12. I. ___II___ II. ___IV___
13. ____metaphase____
14. ___a___
15. ___b___



Part 4: Protein Synthesis

General Knowledge

1. ___d___
2. ___c___
3. ___a___
4. ___a___
5. ___b___
6. ___b___
7. ___a___
8. ___c___
9. ___c___
10. ___b___

Challenge Problems

11. ___e___
12. ___c___
13. ___2400___
14. ___Ala or alanine___
15. ___d___

Scores	
Part 1: _____ / 30 Part 2: _____ / 30 Part 3: _____ / 20 Part 4: _____ / 20	Total: _____ / 100 Rank: _____

Gopher Invitational - Repaired practice edition - Answer key

Division B | Heredity | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (2 points) Authored written substitute for unavailable source material: Which base pairs with guanine in DNA?

Unit: DNA structure, replication, and mutation.
Worked-case data: A DNA template strand reads 3′-TAC GGA TTC-5′. A mutation changes the third codon’s template to ATC. Classify the mutation.
Separate investigation observations:
Normal mRNA: 5′-AUG GCU UGG AAA-3′ codes Met-Ala-Trp-Lys.
Patient mRNA: 5′-AUG GCU UGA AAA-3′.
Protein size: The patient’s protein is much shorter than normal.

A. Adenine

B. Thymine

C. Uracil

D. Cytosine

Correct: D